

VAMSI KRISHNA UNNAM

vk86294140@gmail.com | +1 (551) 344-6817 | New Jersey, USA
github.com/vk86294140-cloud | linkedin.com/in/vamsi-k31245

PROFESSIONAL SUMMARY

Software Engineer with 3+ years designing, developing, and deploying machine learning systems, end-to-end ML infrastructure and pipelines, and cloud-native applications across financial services and autonomous vehicles. Strong in Python, PyTorch, TensorFlow, and AWS, with hands-on expertise in LLM integration, agentic AI workflows, deep learning, and large-scale data engineering.

TECHNICAL SKILLS

ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn, LightGBM, OpenCV, CNN, U-Net, DeepLab

AI & LLM

EXPERIENCE

Software Engineer - Capital One

Jul 2024 - Present

- Built an AI-powered incident-summarization feature end to end - REST backend, LLM inference, context injection, and structured rendering - with iterative prompt tuning across diverse incident types.
- Developed agentic AI workflows orchestrating LLM-driven agents in parallel for incident triage and analysis.
- Built shared validation tooling and reusable test harnesses for ML pipelines, reducing regression-detection time in production.
- Implemented automated failover and chaos-engineering experiments (Gremlin) across AWS for multi-region resilience.

Software Engineer - Tata Consultancy Services

Jul 2021 - Aug 2022

- Engineered deep-learning models (U-Net, DeepLab) for autonomous-vehicle perception - lane, boundary, sign, and object detection - using TensorFlow and Python.
- Designed scalable semantic-segmentation pipelines with TFRecord serialization, generating training-ready datasets at scale.
- Containerized ML workflows with Docker and Kubernetes; tuned and cross-validated models for accuracy.

PROJECTS

PromptPress - LLM Context Compression

Zero-dependency Python library cutting LLM input-token costs 30-70% via a budget-driven compression pipeline; drop-in Anthropic SDK middleware, 29-test suite, CI.

Credit-Risk MLOps Pipeline

End-to-end reproducible pipeline that trains, versions, and serves a credit-default model behind a schema-validated FastAPI service with a CI test matrix.

Document RAG Assistant

Retrieval-augmented generation service with source-grounded citations - chunking, embeddings, cosine-similarity vector store, Claude generation, fully offline-testable.

EDUCATION

M.S. Computer Science (GPA 3.5/4.0) - Stevens Institute of Technology, Hoboken, NJ

2022 - 2024

B.E. Electronics & Communication (CGPA 8.7/10) - Anna University, India

2017 - 2021